

Team Project Proposal

Project Title:

The Correlation Between Video Games' Recognitions in the Capital Market and the Female Fandom Community

Partner:

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Project Description:

This project investigates the relationship between a video game's commercial performance and its reception among female audiences. Using revenue and sales data from Sensor Tower as a measure of commercial success, and AO3 fanfiction creation volume as a proxy for female audience engagement, we analyze whether games that perform well in the broader market also resonate with female players.

Our hypothesis is that commercial success and female audience engagement do not always align. A game can dominate sales charts while offering little meaningful female representation, and conversely, games with strong female fanbases on AO3 may not rank among top commercial performers. By mapping these two dimensions across a range of game titles and genres, we aim to surface patterns that reveal the scale of an underserved market.

People who can benefit from this analysis:

1. Female players seeking games suited to their tastes.
2. Female players who wish to be seen (e.g., in the face of claims by male players that games cater exclusively to men).
3. Game developers who strive to accommodate the needs of audiences of all genders.

Insights we derive from the data:

1. Game popularity in the general market and the female fandom community largely move in tandem. The gender filter bubble effect, while prevalent in many areas of digital media, appears relatively limited in the gaming market.
2. Most video games have a substantial female audience, even games that are traditionally categorized as "male-oriented" (such as action, shooter, and crime-themed titles) have a significant female player base that is willing to pay for them.
3. Commercially successful games do not necessarily treat female characters with respect or align with the psychological needs of female players. Consequently, the full potential of the female gaming market remains an area that warrants greater attention and further development.

Checking the goodness of the analytic:

1. Data Validity:

The data sources utilized in this study possess a high degree of reliability and representativeness. First, commercial performance data was obtained from Sensor Tower (a platform widely used in mobile app market analytics) which provides structured data including metrics such as download volume, revenue estimates, and market performance indicators. As this data is derived through multi-source collection and statistical modeling, it serves as a reasonably accurate reflection of a game's performance at a commercial level.

Second, the use of AO3 ([Archive of Our Own^{beta}](#)) fanfiction creation volume as a proxy indicator for female audience engagement is supported by existing literature. Rouse and Stanfill (2025) surveyed 4,874 AO3 users and found that the platform's user base consists predominantly of cisgender women. Users' identity significantly influences the content they

create and their platform usage behaviors. This finding suggests that the volume of fanfiction written for a specific game on AO3 reflects how much that game resonates with female audiences, and not just general player activity

Game launch dates and version milestones are used as alignment anchors, and data is aggregated on a weekly or monthly basis to ensure comparability across sources.

2. Method Validity:

This study employs correlation analysis to assess the degree of association between a game's commercial performance metrics (Steam revenue and sales volume) and indicators of female audience engagement (AO3 fanfiction creation volume). This cross-dataset comparative approach is well-precedented in the fields of media and consumer behavior research. For instance, in their study on the monetization behaviors of female mobile gamers, Teng et al. (2024) similarly combined commercial metrics with user behavior data to analyze consumption motivations within the female market, thereby validating the methodological soundness of such cross-metric analytical frameworks for studying the female gaming market. Moreover, the use of user-generated content (such as fanfiction) as a quantitative metric for audience emotional engagement is a widely adopted practice in the fields of fan studies and media research, effectively capturing the depth of audience involvement in ways that commercial sales data alone cannot reflect.

3. Robustness of Results:

To ensure the robustness and reproducibility of the analytical findings, this study validates its results across multiple dimensions. First, regarding the specification of time windows, we align AO3 creative activity with commercial gaming metrics using various levels of granularity (e.g., ± 7 days, ± 14 days, monthly aggregation). This approach allows us to test whether our conclusions are sensitive to the specific temporal partitioning, thereby avoiding biases that might arise from the selection of a particular time window.

Second, concerning indicator selection, we do not rely solely on a single commercial performance metric (such as revenue). We will comprehensively incorporate multi-dimensional indicators, including download volume, rankings, and awards, into our analysis. Furthermore, we cross-validate these findings against AO3 data using different statistical measures (e.g., total number of works, tag frequency) to ensure that our results are not contingent upon the definition of any specific indicator.

Additionally, we conduct subgroup analyses across different types of games (e.g., categorized by genre or target audience) to verify the consistency of our conclusions across various subsamples, thereby ruling out the potential influence of individual outlier cases on the overall results. We also employ comparative analysis and outlier detection techniques to control for potential external confounding factors, such as major events or cross-media promotional campaigns.

Finally, the analytical workflow of this study is grounded in standardized data processing and statistical methodologies. All procedural steps are fully reproducible, yielding consistent results under identical data conditions; this further guarantees the stability and reliability of the study's conclusions.

4. References:

(Shared by Jingjing)

J. Gao, C. Xie and C. Tao. (2016). *Big Data Validation and Quality Assurance -- Issues, Challenges, and Needs*. 2016 IEEE Symposium on Service-Oriented System Engineering (SOSE), Oxford, UK, 2016, pp. 433-441

<https://ieeexplore.ieee.org/abstract/document/7473058/authors#authors>

C. Huang, S. Zhu and D. Wu. (2012). *Towards Trusted Services: Result Verification Schemes for MapReduce*. 2012 12th IEEE/ACM International Symposium on Cluster, Cloud and Grid Computing (ccgrid 2012), Ottawa, ON, Canada, 2012, pp. 41-48

<https://ieeexplore.ieee.org/abstract/document/6217403>

(Shared by Celia)

Teng, Z., Wu, X., Chen, C., & Nak-Hwan, C. (2024). *What Makes Female Players Pay for Female Mobile Games?* *Sage Open*, 14(3).

<https://journals.sagepub.com/doi/10.1177/21582440241265322>

Rouse, L., & Stanfill, M. (2025). *Quantifying fandom: Identity, romance, and platform preferences on archive of our own*. *New Media & Society*, 0(0).

<https://journals.sagepub.com/doi/10.1177/14614448251367033>

Data sources:

1. (shared by Celia)

AO3 Official Dataset: https://archiveofourown.org/admin_posts/18804

Data type : String, Integer

Schema:

The first csv includes information about works:

Field	Data Type	Description
creation_date	DATE	Date the work was published
language	STRING	Language the work is written in
word_count	INTEGER	Total word count of the work
restricted	BOOLEAN	Whether the work is restricted to logged-in users
complete	BOOLEAN	Whether the work is marked as complete
tag_id	INTEGER	IDs of tags associated with the work

The second csv provides the key to the tag IDs:

Field	Data Type	Description
tag_id	INTEGER	Unique identifier for each tag
tag_type	STRING	Category of tag (e.g. Fandom, Relationship, Warning)
tag_name	STRING	Name of the tag (omitted if fewer than 5 uses)
canonical	BOOLEAN	Whether the tag is a canonical/official tag
uses	INTEGER	Approximate number of times the tag has been used
merger_id	INTEGER	ID of the canonical tag this tag merges into, if applicable

2. (shared by Jingjing)

Sensor Tower Games Database:

<https://app.sensortower.com/vgi/games-database>

Data type: String, Float, Integer

Schema:

Current performance of all Steam games:

- Revenue in \$

- Units sold

- Total followers

- Follower gain in the last 2 weeks

- 24h active players peak

- Rating

Historical performance:

-Revenue, units sold, price, reviews and followers by date

-Daily concurrent players: peak, average, median and minimum number of players per day.

Price history in all supported currencies:

-Price changes of a Steam game in all the currencies in which the game is sold on Steam.

Game ranking among other Steam games:

-Lifetime reviews, lifetime revenue, lifetime units sold, units sold yesterday, lifetime followers, 24h active players, average playtime global rankings (eg 150 / 75,000)

Player insights for all Steam games with >500 reviews:

-Average playtime of the game

-Average playtime in hours

-Units sold by country for the 20 largest countries

-Country name and code

-Percentage split

-Units sold by region for all regions

-Region name

-Percentage split

-Game overview

-Game price

-Release date

-Game genres, subgenres

- Game mode, theme, art style, character
- Steam tags
- Release type
- Supported languages
- Publisher classification (AAA / AA / Indie)
- Publishing
- List of developers and publishers

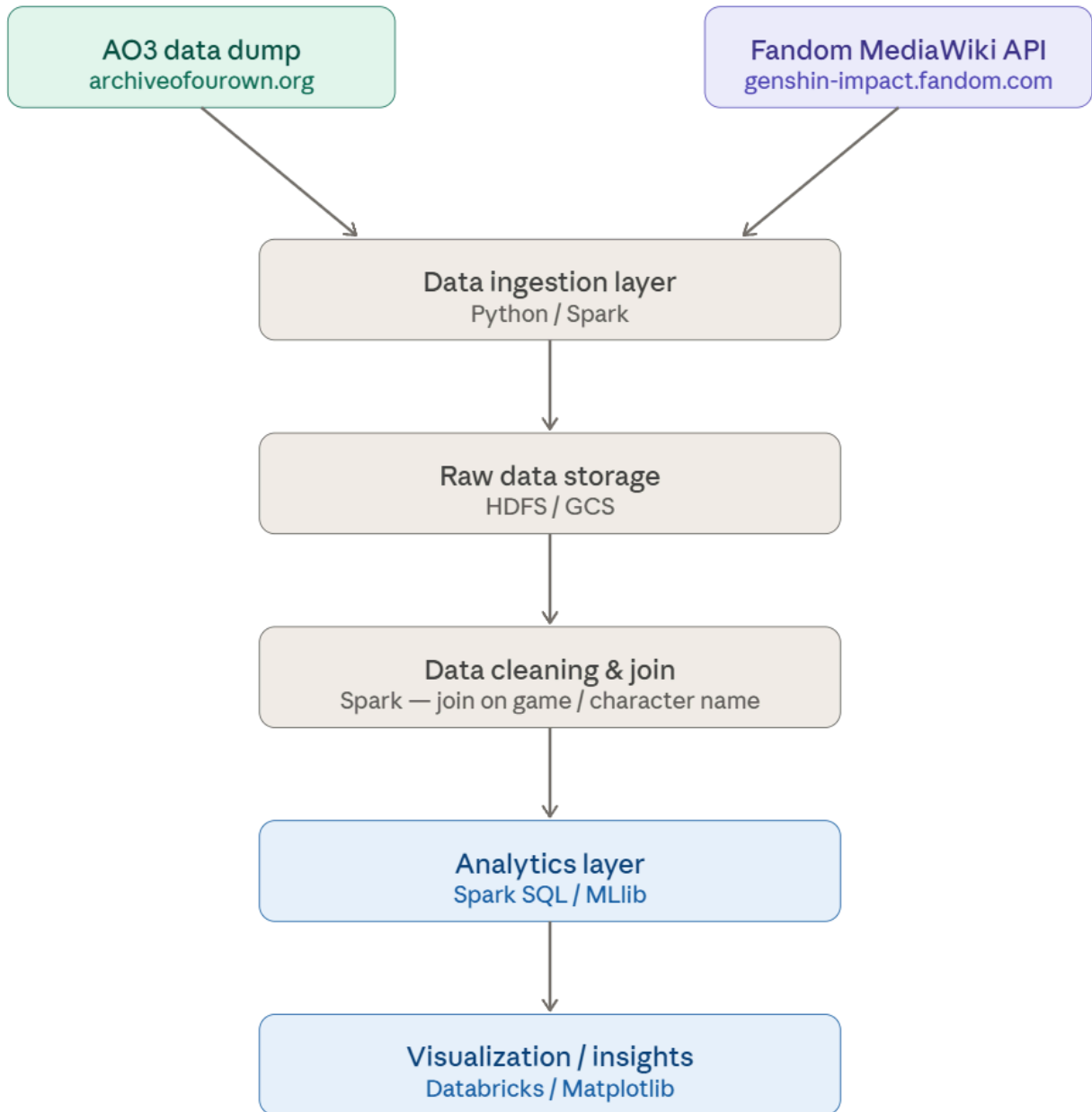
Company overview:

- Name
- Classification (AAA / AA / Indie)
- Games developed or published
- Games in development
- Total revenue in \$
- Average and median revenue per game in \$

Company lists:

- Lists of all Steam developers and publishers in VGI's database. The list includes the ID and name of each company.

Initial design diagram:



Task List in Click up:

Rough plan for the whole project:

Team Space / Rough Plan

Add Channel

List

Board

Calendar

Gantt

Table

+ View

Group: Status

Subtasks

Columns

Filter

Closed

Assignee

J

Q

+

Add Task

▼

TO DO

9

Name	Assignee	Due date	Priority	Status	Comments	+
Create Team Project Proposal.	JW C	Tomorrow	High	TO DO		
Data Acquisition & Ingestion	JW C	Tue	Normal	TO DO		
Data Cleaning & Preprocessing	JW C	4/4/26		TO DO		
Data Profiling & EDA	JW C	4/8/26		TO DO		
Initial Analytics Development	JW C	4/13/26		TO DO		
Advanced Analytics & Optimization	JW C	4/17/26		TO DO		
Evaluation & Validation	JW C	4/21/26		TO DO		
Visualization & Insight Generation	JW C	4/25/26		TO DO		
Final Code & Report Submission	JW C	4/27/26		TO DO		
+ Add Task						
+ New status						

Task list for milestone 1:

Team Space / Create Team Project Proposal

Add Channel

List

Board

Calendar

Gantt

Table

+ View

Group: Status

Subtasks

Columns

J

Q

+

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IN PROGRESS

7

Name	Assignee	Due date	Priority	Status	Comments	+
Project Discription		Today	Urgent	IN PROGRE...		
Who benefits from the analysis		Tomorrow		IN PROGRE...		
Insight from the data		Tomorrow		IN PROGRE...		
Check the goodness		Tomorrow		IN PROGRE...		
Find data sources		Tomorrow		IN PROGRE...		
Draw initial design diagrams		Tomorrow		IN PROGRE...		
Create task list	JW	Today	Urgent	IN PROGRE...		
+ Add Task						

Home

+ Create

Inbox

Replies

Assigned Comments

My Tasks

More

Favorites

Add to your sidebar

Spaces

All Tasks - Processing Big D...

Team Space

Rough Plan

Create Team Project ... 2

Data Profiling & Inge... 5

Team Docs

New Space

Channels

General - Processing Big Data

Welcome

Add Channel

Direct Messages

Jingjing Wang — You

New message

ClickUp needs your permission to send notifications

Team Space / Create Team Project Proposal

Add Channel

List

Board

Calendar

Gantt

Table

+ View

Group: Status

Subtasks

2 collapsed

COMPLETE 7

Project Discription

JW

L

Yesterday

Urgent

Who benefits from the analysis

JW

L

Today

Insight from the data

JW

L

Today

Check the goodness

JW

L

Today

Find data sources

JW

L

Today

Draw initial design diagrams

JW

L

Today

Create task list

JW

Yesterday

Urgent

+ Add group